

Machine Learning Based Mental Health Prediction System for Early Detection of Depression and Stress Among Students

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Abstract

This paper presents a Machine Learning based system for predicting mental health conditions such as stress and depression among students. The proposed system analyzes behavioral and lifestyle attributes including sleep patterns, academic pressure, mood swings, and social interaction. Machine Learning algorithms including Logistic Regression, Decision Tree, and Random Forest were implemented and compared. Experimental analysis showed that the Random Forest model achieved the highest prediction accuracy and produced reliable results for early detection. The developed system can support students, educational institutions, and healthcare professionals by providing faster and more accessible mental health analysis.

Introduction

Mental health problems are increasing rapidly among students due to academic stress, career uncertainty, and lifestyle imbalance. Depression and anxiety directly affect concentration, academic performance, and emotional well-being. Traditional diagnosis methods require significant time and expert consultation, which may not always be accessible. Machine Learning offers an intelligent and scalable solution capable of analyzing large datasets and identifying mental health risks at an early stage. The objective of this project is to build an automated prediction system that assists in identifying stress and depression levels using behavioral data.

Objectives

- Develop a Machine Learning based mental health prediction system.
- Analyze behavioral and psychological attributes related to stress and depression.
- Compare multiple Machine Learning algorithms for prediction accuracy.
- Provide faster and automated prediction results.
- Create a simple and user-friendly interface for users.

Literature Review

Several researchers have applied Machine Learning techniques in healthcare and mental health analysis. Previous studies used Logistic Regression, Support Vector Machine, and Random Forest algorithms for predicting depression and anxiety levels. Some systems achieved good accuracy but faced limitations such as poor scalability, low real-time performance, and limited datasets. Recent advancements in Machine Learning and Artificial Intelligence have improved prediction efficiency and model reliability in healthcare applications.

Problem Statement

Mental health disorders are difficult to identify during their early stages because many individuals avoid seeking professional support. Traditional methods require manual assessment and significant resources. There is a need for an intelligent system capable of automatically predicting mental health conditions using user behavioral data and Machine Learning techniques.

Proposed System

The proposed system collects user information through questionnaires related to sleep habits, social interaction, academic pressure, mood swings, and anxiety levels. The collected data is preprocessed and passed through Machine Learning algorithms for prediction. The system then generates prediction results and displays them through a user-friendly dashboard.

Dataset and Preprocessing

The dataset used in this project was collected from publicly available mental health datasets available on Kaggle. Important attributes include age, gender, sleep hours, work pressure, anxiety level, lifestyle habits, and mental fatigue. Data preprocessing steps included removing null values, eliminating duplicate records, encoding categorical variables, feature scaling, and normalization. These preprocessing techniques improved model performance and prediction accuracy.

Algorithms Used

Logistic Regression: Used for binary classification problems and simple predictive analysis.

Decision Tree: Generates classification rules based on feature importance and decision conditions.

Random Forest: An ensemble learning algorithm that combines multiple decision trees to improve accuracy and reduce overfitting. This algorithm achieved the highest prediction accuracy in the proposed system.

Implementation

The frontend was developed using HTML, CSS, and JavaScript. The backend implementation was completed using Python Flask. Machine Learning model training and evaluation were performed using Python libraries such as Pandas, NumPy, Scikit-learn, and Matplotlib.

Results and Discussion

The system was evaluated using multiple Machine Learning algorithms. Random Forest achieved the highest accuracy due to its ensemble learning capability and better handling of complex datasets. Evaluation metrics included Accuracy, Precision, Recall, and F1-Score.

Future Scope

Future improvements may include Deep Learning integration, AI chatbot support for counseling, mobile application development, cloud deployment, wearable device integration, and real-time emotion analysis.

Conclusion

The proposed Machine Learning based Mental Health Prediction System successfully predicts stress and depression levels using behavioral and psychological attributes. Experimental analysis showed that Random Forest provided the best performance with high prediction accuracy. The system provides an efficient and scalable solution for early mental health prediction and healthcare assistance.

Performance Comparison

Algorithm	Accuracy
Logistic Regression	81%
Decision Tree	85%
Random Forest	91%

References

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